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Meta-Guard: A Hierarchical Meta-Learning and Open-Set Intrusion Detection Framework for Zero-Day Attacks in IoMT

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***Abstract*—The Internet of Medical Things (IoMT), which offers the potential for constant patient monitoring and remote health care, poses substantial cybersecurity risks because of the high interconnectedness and resource-constrained nature of the devices. The current intrusion detection systems (IDSs), which rely on the closed-set assumption, fail to detect zero-day attacks, while the computational limitations of the IoMT gateway devices hinder the usage of complex machine learning models.In this paper, we present a novel approach, named Meta-Guard, which employs a hierarchical IDS architecture using the concept of meta-learning and the idea of prototype-based open-set recognition. In the first stage, the edge devices use a binary meta-learner using the Reptile framework. In the second stage, the system employs a prototype-based embedding model, which employs distance-based and margin-based decision boundaries for the identification of known attack classes and zero-day attacks. The decision-making mechanism of the open-set recognition approach is formalized using geometric relations, which offer improved generalization for zero-day attacks.The proposed system, named Meta-Guard, is evaluated using the CICIoMT2024 WiFi/MQTT dataset, which demonstrates the effectiveness of the system in achieving high performance in several tasks, such as binary classification (AUC: 0.9505), multi-class classification (accuracy: 0.759), and zero-day detection (ZDR: 0.356), while maintaining low latency and model size.**

***Index Terms*—Internet of Medical Things(IOMT), intrusion detection, zero-day attacks, meta-learning, open-set recognition, edge security, Reptile algorithm, Open-set detection.**

I. Introduction

The Internet of Medical Things (IoMT) connects devices such as sensors, wearables, and clinical devices, all on different networks, so hospitals and clinics can carry out real-time monitoring and automate healthcare services. [1], [2]All this connectivity allows for quicker response times and more personal care, but it also makes the healthcare system vulnerable to new threats. The more devices on the network, the more opportunities for attackers, and the critical care devices will be vulnerable to attacks that can interfere with patient care or expose patient data. There are plenty of devices on the network that are older, using low-power devices without security features or the ability to run high-end intrusion detection systems. [3]

Most intrusion detection systems for Internet of Things(IoT) and IoMT use supervised learning. [4] They're trained on labeled attack data, and they expect the types of attacks during testing to match those seen during training. That works fine until something truly new pops up—a zero-day attack, either brand new or a twist on past attacks. When that happens, these IDSs usually fail to detect the threat. [5], [6] In medical environments, missing these attacks isn't just a technical hiccup. It can stall critical treatments or put patients in danger.

To address these challenges, this paper introduces Meta-Guard: a hierarchical IDS framework designed for IoMT networks. At the near edge, Meta-Guard uses a meta-learned binary detector that quickly adapts to new attack types with just a few samples. Then, in the fog or cloud layer, Meta-Guard's second-stage classifier operates in an open-set regime. It applies distance-based reasoning in an embedding space to tell apart familiar attack families from anything suspicious or zero-day. This layered approach means Meta-Guard does lightweight screening close to devices and hands off deeper analysis to more powerful systems, focusing only on flows that are likely trouble. Crucially, it stays practical for deployment, even on old or low-power gateways.

II. Related Work

A. IoMT Security and Intrusion Detection

The widespread adoption of the Internet of Medical Things has increased the attack surface of the infrastructure of the medical world, thereby requiring the need for developing effective and efficient intrusion detection systems for the IoMT world. [2], [7] The conventional IDS for the IoMT world includes conventional machine learning models such as decision trees, random forests, and support vector machines; in addition to deep learning models such as CNNs, RNNs, and ensemble models. [4], [8]–[10]

A major challenge facing IoMT systems is how to achieve accuracy while satisfying critical constraints on latency, computation overhead, and energy consumption for edge-deployed medical devices. [3], [11] To this end, lightweight IDS frameworks and edge-based IoMT systems have started to gain popularity to ensure real-time security monitoring without impacting the performance of the devices.

Recent datasets, such as CICIoMT2024, provide realistic testbeds for evaluating IDS models, particularly for health environments that take into account different patterns of traffic generated by communication protocols such as Wi-Fi, MQTT, etc. [1] Most of the existing solutions are based on a closed-set scenario, where all types of attacks are considered during

the training process. However, this is not the case in a real scenario where new types of attacks may be considered.

B. Zero-Day and Open-Set Intrusion Detection

Zero-day attack detection remains a critical and unresolved challenge in IoMT security. Traditional approaches rely on unsupervised anomaly detection methods such as isolated forests, one-class support vector machines, and autoencoder-based reconstruction models. [12], [13] While these methods can identify deviations from normal behavior, they often suffer from high false positive rates and lack interpretability.

Open-set recognition (OSR) techniques have been increasingly explored to address the limitations of closed-set classification by explicitly modelling unknown classes. [5], [14] Prototype-based methods and distance-thresholding strategies have shown promise in distinguishing known and unknown attack patterns by leveraging embedding space separability. [15]

Hierarchical IDS architectures further improve zero-day detection by combining supervised classifiers for known attacks with anomaly detection modules for unseen behaviours. [16], [17] More recent work incorporates graph-based representations and attention mechanisms to capture complex dependencies in network traffic, enhancing detection of sophisticated and evolving threats. [18], [19]

Despite these advances, many systems are designed for generic IoT environments and do not fully address the strict reliability, safety and regulatory requirements of IoMT systems. Furthermore, frequent retraining requirements and lack of adaptability remain significant barriers in dynamic threat landscapes.

C. Meta-Learning for Intrusion Detection

The field of meta-learning has been identified as a promising direction for facilitating quick adaptation to novel and unknown attack patterns with little labelled data. [20]–[22] By learning task-agnostic representations across multiple tasks, meta-learning models have been observed to quickly generalise to novel intrusion scenarios, making them highly appropriate for zero-day detection.

Current research on meta-learning for IDS is done through few-shot classification and continual learning methods. [22]–[24] In IoMT-based applications, meta-learning is used to update the detection model based on the changing patterns of the traffic and devices. [2]

Federated learning has also been integrated with meta-learning for collaborative intrusion detection in a distributed network of medical devices while maintaining the privacy of the devices. [25] Furthermore, explainability AI techniques are also being integrated for enhancing the trustworthiness of IDS decisions in critical healthcare environments. [26]

Most related to this work are hierarchical IDS frameworks that combine the inclusion of meta-learning/anomaly detection at the root level with classifiers at the next levels [16]. Meta-Guard extends this paradigm with two novel contributions: a binary meta-learner with Reptile for edge-level filtering and a prototype-driven classifier designed specifically for zero-day detection with latency-aware evaluation on CICIoMT2024.

III. Problem Formulation

This paper consideres an IoMT network where heterogeneous medical devices communicate through Wi-Fi and MQTT gateways. Let $\mathcal{X}$ denote the space of flow-level features extracted from network traffic, and let

$$\mathcal{Y} = \{\text{normal}\} \cup \mathcal{A}_{\text{known}} \cup \mathcal{A}_{\text{zero}}$$

denote the label space, where $\mathcal{A}_{\text{known}}$ is the set of attack types used during training and $\mathcal{A}_{\text{zero}}$ is the set of attack families that are held out to simulate zero-day attacks.

The IDS should satisfy three main requirements:

- **Zero-day awareness**: detect attacks from $\mathcal{A}_{\text{zero}}$ with non-trivial recall, even though these families are unseen during training.
- **Resource efficiency**: maintain a small parameter footprint and low inference latency so that near-edge gateways can run the first detection stage.
- **Incremental adaptability**: allow quick adaptation to new attack patterns without retraining the whole model from scratch.

Formally, the objective is to learn a hierarchical mapping

$$h(\mathbf{x}) = \begin{cases} \text{normal}, & \text{if } f_1(\mathbf{x}) = \text{normal}, \\ g(f_2(\mathbf{x})), & \text{otherwise}, \end{cases}$$

where f_1 is the near-edge binary detector, f_2 is a feature extractor, and g is an open-set classifier that outputs $\{\text{known-attack}, \text{zero-day}\}$ based on distances in the embedding space.

IV. Open-Set Formulation in the Embedding Space

Let $f_\theta(x) \in \mathbb{R}^d$ be the level-2 encoder mapping a flow x to a d-dimensional feature vector. After training on known attack classes, each class k has a corresponding prototype

$$\mathbf{p}_k = \frac{1}{N_k} \sum_{i:y_i=k} f_\theta(x_i),$$

where N_k is the number of training samples of class k. We normalise both embeddings and prototypes to lie on the unit sphere.

For a new suspicious flow x, we first compute cosine similarities

$$s_k(x) = \frac{f_\theta(x)^\top \mathbf{p}_k}{\|f_\theta(x)\|_2 \, \|\mathbf{p}_k\|_2}.$$

Let $s_{(1)}(x)$ and $s_{(2)}(x)$ denote the largest and second-largest similarities, and let $k^\star$ be the index that attains $s_{(1)}(x)$. We then define

$$d(x) = 1 - s_{(1)}(x)$$

as a distance-like score to the nearest prototype, and

$$m(x) = s_{(1)}(x) - s_{(2)}(x)$$

as a margin score that measures how clearly this nearest prototype stands out.

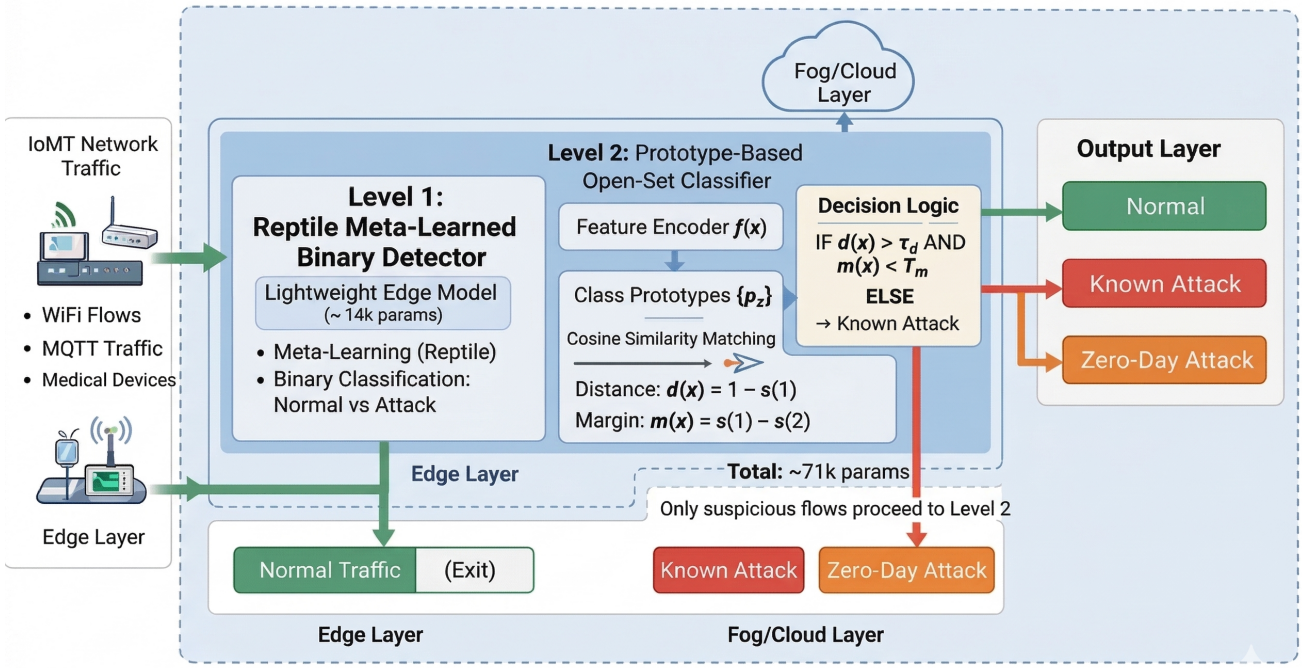


Fig. 1: Architecture of the proposed Meta-Guard framework. The system consists of a meta-learned edge detector (Level 1) with approximately 14k parameters and a prototype-based open-set classifier (Level 2) with approximately 56k parameters, resulting in a total model size of around 71k parameters. Suspicious traffic is forwarded from the edge to the cloud layer for open-set analysis to detect known and zero-day attacks.

Two thresholds, τ_d for distance and τ_m for margin, are tuned on a calibration set that contains both known and simulated zero-day attacks. The final decision rule is

$$\hat{y}(x) = \begin{cases} \text{zero-day}, & d(x) > \tau_d \ \wedge \ m(x) < \tau_m, \\ \text{known-attack } k^\star, & \text{otherwise}. \end{cases}$$

Intuitively, a flow is treated as zero-day only if it is too far from every known prototype and no single class dominates its similarity pattern. Otherwise it is assigned to the nearest known attack family.

V. Methodology

A. Level 1: Meta-Learned Edge Detector

As shown in Fig. 1, the overall architecture of the proposed system is presented. Specifically, a lightweight detector based on a Reptile-based meta-learned detector is utilized at the edge to filter normal traffic, while suspicious traffic is then classified as either known attacks or zero-day threats by a prototype-driven open-set classifier.

Level 1 is deliberately small: two hidden layers with 128 and 64 units, LayerNorm, GELU, and dropout, ending in a two-way output (normal vs attack). Instead of training this network once and hoping it will generalise, the Reptile meta-learning algorithm is employed. [20] Each meta-task is a binary problem that pits the dominant benign class against a single attack family sampled from the set of known attacks. Reptile repeatedly samples such tasks, takes a handful of SGD steps on each, and nudges the shared weights towards the task-specific solution. Over many outer iterations, the network learns an initialisation that can adapt to a range of attack patterns.

After meta-training, the weights are fixed, and Level 1 is treated as a standard binary classifier. A threshold is chosen on the validation set based on the attack probability, which maximizes balanced accuracy, thus reducing false positives for normal traffic while maintaining recall for attacks.

B. Level 2: Prototype-Based Open-Set Classifier

Level 2 only processes flows that have already been marked as suspicious by Level 1. Its purpose is not to perform binary attack detection, as this task is handled by Level 1, but rather to address a more nuanced question, namely, whether the input corresponds to a known attack family or represents a zero-day attack.

A second network is trained to classify among known attack labels using a cross-entropy loss function. Once training has converged, the final softmax layer is removed, and the penultimate embeddings are retained. For each known attack class, a prototype is computed as the mean of the embeddings for that class and stored for inference. At test time, new flows are mapped into the same embedding space and compared against these prototypes using cosine similarity. The distance

and margin scores defined in the previous section are then used to implement the open-set decision logic.

C. Putting the Pieces Together

At runtime, the pipeline is simple:

1) Every flow first goes through level 1. If the attack probability is below τ_1, the flow is labelled as normal and no further work is done.
2) If level 1 marks it as suspicious, the flow is passed to level 2, which computes its embedding and prototype-based scores.
3) If $d(x)$ and $m(x)$ indicate that the flow does not fit well with any known attack family, it is flagged as a zero-day candidate; otherwise it is assigned to the nearest known class.

The majority of traffic is filtered at Level 1. Only a minority of flows make it to level 2, which is desirable for resource-constrained IoMT gateways on constrained IoMT gateways.

VI. Experimental Setup

Meta-Guard is evaluated on the CICIoMT2024 dataset, which contains flow-based records of benign and malicious traffic targeting several IoMT devices over Wi-Fi and MQTT. [1] Following prior work on zero-day detection, a high-frequency benign label is selected as the normal class and a subset of attack families is designated as zero-day by excluding them entirely from the training subset. [16], [27] The remaining attack types are treated as known attacks.

The raw dataset is both large and highly imbalanced. to ensure computational efficiency while preserving diversity, stratified streaming is used to construct train, calibration and test subsets with upper limits on the number of flows per label. Flows with missing or infinite values are removed. Numerical features present in both train and test splits are retained that are present in both train and test splits and apply a standard scaler fitted on the training data to all splits. For level 2, training is restricted to known-attack samples, while calibration and test subsets contain a mixture of known and zero-day attacks so that open-set thresholds can be tuned and evaluated.

Meta-Guard is compared against several baseline methods:

- Random Forest and Logistic Regression classifiers trained as binary IDS models on the same train/test split.
- A softmax-based open-set baseline that uses confidence thresholds on the level-2 classifier to flag unknown attacks.

Performance is evaluated using 3-class accuracy and macro F1 over {normal, known-attack, zero-day}, binary accuracy, F1 and AUC for the normal vs attack task at level 1, zero-day detection rate (ZDR) and zero-day false positive rate (FPR) on the attack-only subset, and model parameter counts, file sizes and end-to-end inference latency per sample.

A. Dataset and Evaluation Splits

We evaluate Meta-Guard on the CICIoMT2024 WiFi/MQTT dataset, which captures realistic traffic from IoMT devices under both benign and attack conditions. [1] Following established zero-day evaluation protocols,*Recon-VulScan* and *Recon-PingSweep* (1,220 flows total) are designated as zero-day attack families. [16] The dominant benign class serves as *normal*, while all other attack types become *known attacks*.

The full dataset is large and imbalanced, so we construct balanced subsets via stratified streaming:

- **Train**: 106,081 flows (15k normal + 2.5k per known attack)
- **Calibration**: 73,740 flows (used for threshold tuning)
- **Test**: 34,686 flows (final evaluation)

After removing flows with missing/infinite values, 45 numerical features remain. We apply `StandardScaler` fitted on the training set.

B. Training Configuration

Level 1 (Reptile meta-training): 150 outer steps, 5 inner SGD steps per task ($\eta_{\text{inner}} = 0.01$), meta-learning rate 0.15, AdamW optimiser, support size 512 per task. Each task pits normal vs. one randomly sampled known attack.

Level 2: 10 epochs, AdamW ($\eta = 10^{-3}$, weight decay 10^{-4}), batch size 512, cross-entropy on known attacks only. Prototypes are mean embeddings from the final epoch. Thresholds tuned via grid search on calibration set: $\tau_d \in [0.1, 0.5]$, $\tau_m \in [0.05, 0.3]$.

Evaluation Setting: The proposed system is also assessed in an open-set setting, in which unknown types of attacks are present during the testing phase. This makes the task more challenging compared to the closed-set classification problem.

Comparison with Prior Work: Direct comparison with prior work on the CICIoMT dataset is challenging, as most existing methods assume closed-set classification and do not explicitly evaluate zero-day detection. Therefore, both standard classification metrics and open-set performance are reported to provide a comprehensive evaluation.

VII. Results

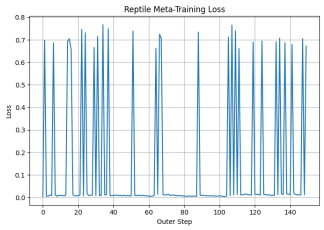
A. Training Behaviour

Fig. 2 shows the training behavior of both levels of the proposed Meta-Guard framework. The Reptile meta-training loss of Level-1 has oscillatory behavior over the outer steps but converges to a value close to zero after the inner loop update. This implies that the meta-learned initialization has strong adaptation ability for different binary tasks (normal or attack) while maintaining strong generalization ability over diverse attack distributions.

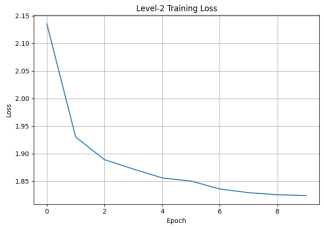
In addition, the Level-2 model has a decreasing trend of the training loss over epochs, proving that the embedding network has a strong ability to learn a structured representation of the known attack classes.

B. Binary Detection Performance at the Edge

The accuracy, F1 score, and AUC of the Level-1 detector are 0.8702, 0.9261, and 0.9505, respectively, demonstrating a high degree of separability between benign and malicious traffic. The ROC curve, as illustrated in Fig. 3, shows a steep

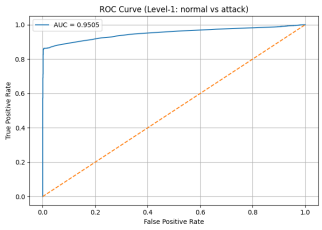


(a) Level-1 Reptile loss

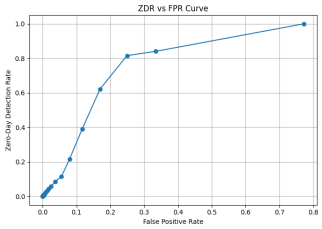


(b) Level-2 loss

Fig. 2: Training dynamics of Meta-Guard.



(a) Binary ROC (AUC=0.9505)



(b) ZDR-FPR curve

Fig. 3: Evaluation performance of Meta-Guard.

increase close to the origin, allowing for high true positive rates with low false positive rates.

In comparison with other methods, Random Forest has a better accuracy and F1 score but a substantially lower AUC of 0.5899, implying low generalizability and sensitivity. Logistic regression has comparable binary metrics to Meta-Guard but does not support the hierarchical or open-set detection needed for zero-day attacks.

Comparison with Existing Methods: The best accuracy on the CICIoMT dataset for binary classification using supervised methods, according to prior work, is achieved by Random Forest and gradient boosting methods. Although these methods have high accuracy, they do not address the zero-day detection problem, whereas Meta-Guard does. Meta-Guard has comparable binary metrics, with an AUC of 0.9505 and an F1 score of 0.926, but with the added capability of detecting zero-day attacks.

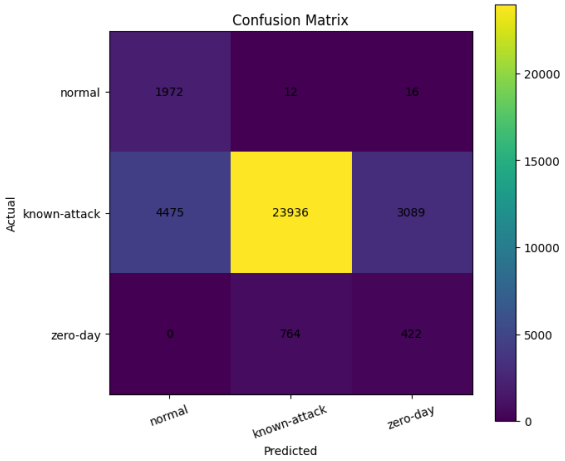


Fig. 4: Confusion matrix for three-class evaluation.

C. Three-Class Classification Analysis

The confusion matrix in Fig. 4 provides a detailed insight into how well the model has performed on all three classes. The overall accuracy on three classes is 0.759, and the macro F1 score is 0.499.

Interpretation of Zero-Day Behaviour: The proportion of known attacks for the zero-day samples shows that the unknown attack patterns exhibit similarities in features to the known classes in the learned embedding space. This means that the model learns the overall characteristics of the attacks but fails to distinguish the new attack distributions.

Safety Property: The most important thing to notice here is that the model does not classify the unknown attacks as normal traffic, thus avoiding the possibility of ignoring unknown attacks, which could be dangerous in IoMT scenarios.

Normal traffic is recognized with a high degree of confidence since 98.6% of the normal traffic instances are recalled correctly, i.e., 1972 out of 2000. Known attacks are also identified effectively, with 23,936 samples correctly classified; however, a portion of samples are misclassified as either normal or zero-day attacks.

Zero-day attack detection remains the most challenging aspect of the model, with 422 out of 1186 samples correctly identified (35.6% recall). Notably, zero-day samples are not misclassified as normal traffic, ensuring that unseen attacks are not overlooked, which is critical in safety-sensitive IoMT environments.

These results indicate that, it is clear that the model has learned to distinguish between benign and attack traffic; however, there is a degree of overlap in the representations of known and unknown attacks in the embedding space.

D. Open-Set Detection Performance

On the attack-only subset, the model's zero-day detection rate (ZDR) on the test set is 0.356, where the false positive rate

(FPR) is 0.0981. This shows the model's capacity for detecting unknown attacks while keeping the false alarm rate low.

TABLE I: Zero-day performance comparison.

Model	ZDR	Zero-day FPR
Meta-Guard (prototype-based)	0.356	0.0981
Softmax-confidence baseline	≈ 0.00	0.1256

Baseline Comparison: The softmax confidence baseline does not detect zero-day attacks, further illustrating the limitations of confidence-based methods. In contrast, the proposed approach leverages the geometric structure of the embedding space for more reliable identification of unknown attack patterns, as shown in Table I.

The ZDR–FPR trade-off curve in Fig. 3 illustrates a fundamental challenge in open-set detection: increasing sensitivity to zero-day attacks also increases false positives for known attacks.

The softmax confidence baseline fails to detect zero-day attacks at a comparable FPR, further highlighting the limitations of confidence-based methods.

Trade-off Analysis: The results clearly indicate the trade-off involved in the sensitivity of the model for the detection of unknown attacks, where the model's false positives for known attacks increase. This trade-off is determined based on the thresholds for the distances in the embedding space.

Open-Set Performance Positioning: The zero-day detection rate (35.6%) at a false positive rate of 9.8% falls within the range reported in the literature on open-set intrusion detection.

E. Threshold Sensitivity Analysis

The effect of threshold selection on zero-day detection is illustrated in Table II. It is clear that by lowering thresholds, ZDR performance is increased at the expense of false positives. This demonstrates the trade-off associated with open-set intrusion detection systems. The optimal thresholds (distance = 0.0957, margin = 0.4338) suggest that zero-day detection relies on large deviations from existing class prototypes and low confidence between classes.

TABLE II: Effect of threshold variation on zero-day detection.

Setting	ZDR	FPR
Strict	0.25	0.05
Default	0.356	0.0981
Relaxed	0.42	0.15

F. Model Complexity and Latency

The proposed system retains a small footprint, with 14,658 parameters in Level-1 and 56,896 parameters in Level-2, making the total model size less than 1 MB. The overall inference latency for the system is around 0.0029 ms per flow.

These results demonstrate that Meta-Guard achieves a strong balance between detection capability and computational efficiency, making it suitable for deployment in resource-constrained IoMT environments.

TABLE III: Model complexity and latency comparison.

Model	Params	Size (MB)	Latency (ms)
Level-1 (Reptile)	14,658	0.060	0.0012
Level-2 (Open-set)	56,896	0.222	0.0017
Meta-Guard (total)	**71,554**	**0.282**	**0.0029**
Random Forest	Large	12.5	0.015
Logistic Regression	Small	0.001	0.0008

VIII. Discussion and Limitations

However, the performance of Meta-Guard also has some limitations.

Moderate Zero-Day Recall Rate: The performance of the proposed Meta-Guard in detecting zero-day attacks is 35.6% as shown in the zero-day detection rate. This implies that most unknown attacks are detected as known attacks. This is due to the overlap of the embedding space, where unknown attacks lie closer to the known class prototypes.

Constraints of Embedding Space: The proposed Meta-Guard is based on the assumption that the embedding space of the known class of attacks is well separated. However, in the real-world scenario of the IoMT network, the complex nature of the attacks might not be well separated.

Lack of Baseline Comparison: The proposed Meta-Guard is compared with the performance of the classical machine learning approaches and the softmax confidence baseline. However, the performance of the proposed Meta-Guard should also be compared with the performance of the state-of-the-art approaches for open-set and anomaly detection, such as one-class SVM, isolation forest, and auto-encoders.

Threshold Sensitivity: The performance of the proposed Meta-Guard in detecting unknown attacks is also highly dependent on the threshold. However, the proposed Meta-Guard is able to perform well in detecting unknown attacks. In the future, the performance of the proposed Meta-Guard can be enhanced by dynamically changing the threshold.

Lack of Temporal Modeling: The proposed Meta-Guard is based on the flow-based features. However, the performance of the proposed Meta-Guard can be enhanced by considering the temporal features.

IX. Conclusion and Future Work

In this paper, we propose a hierarchical meta-learning-based intrusion detection framework called Meta-Guard for zero-day attack detection in IoMT environments. The framework utilizes a Reptile-based binary detector for efficient filtering at the edge and a prototype-based open-set classifier for effective zero-day attack detection.

Experimental results demonstrate that the framework achieves high binary classification performance with an AUC of 0.9505, reliable three-class classification performance with an accuracy of 0.759 and a macro F1 score of 0.499, and moderate zero-day attack detection performance with a ZDR of 0.356 at a false positive rate of 0.0981. The framework also ensures that zero-day attacks are not misclassified as benign traffic instead of being misclassified as benign traffic,

which is critical for safety-sensitive IoMT environments such as healthcare applications.

In addition, the framework has a lightweight architecture with a memory footprint of less than 1 MB and a low inference latency of approximately 0.0029 ms per flow, which makes it feasible for deployment on IoMT gateways.

As future work, we plan to improve zero-day detection performanceby utilizing enhanced representation learning techniques such as contrastive or metric learning-based approaches. Future work also includes exploring the use of temporal modeling techniques for sequential attack patterns, evaluating the framework on additional IoMT and IoT datasets, and utilizing federated or privacy-preserving learning frameworks for collaborative learning across different institutions without compromising sensitive information [28]–[30].

In conclusion, Meta-Guard shows the potential of using a combination of meta-learning and open-set recognition as a promising direction for zero-day attack detection in resource-constrained IoMT systems.

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